

Can You Judge a Movie’s Success by Its Poster?

DS 4002 Case Study — Movie Poster Visual Analytics

You’ve just been hired as a **Visual Marketing Analyst** at a major film studio. Your boss walks in with a bold claim: *“I can tell how well a movie will do at the box office just by looking at its poster.”* She believes that certain visual patterns in movie posters — vibrant color palettes, specific color grading trends like the famous “orange and teal” look, and the number of faces on the poster — are reliable predictors of a film’s worldwide theatrical revenue.

Your mission: Use data science to put this claim to the test.

You have access to a dataset of over **1,000 films** with worldwide box office figures. For each film, you will download its official poster and extract quantitative visual features: the dominant color palette, average brightness, color saturation, the prevalence of “Hollywood-style” orange-and-teal color grading, and the density of human faces. You will then apply statistical modeling techniques — including correlation analysis, multiple linear regression, and a machine learning model — to determine whether any of these visual characteristics are statistically significant predictors of global revenue.

The stakes are real. Studios spend millions on marketing, and the poster is often the first thing a potential audience member sees. If visual features genuinely predict revenue, it could reshape how marketing teams design campaigns. If they don’t, it tells us something equally important about what *actually* drives commercial success in the film industry.

Your Deliverable

Produce a **short analytical report** (3–5 pages) with supporting visualizations and statistical results that answers the research question: *Do the visual characteristics of a movie poster predict its worldwide box office performance?* Your analysis should include exploratory data analysis, at least one inferential statistical model, and a predictive evaluation. All code and data should be reproducible.

Details on the specific requirements, data sources, and evaluation criteria are provided in the **Rubric**.

“The data doesn’t lie — but it might surprise you.”